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6 MAY 1959
SUPERSEDING
MIL-S-13171A(SigC)
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MILITARY SPECIFICATION

SWITCHBOARD, TELEPHONE, MANUAL
(TELEPHONE SWITCHBOARD SB-55( )/FTC)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers Telephone Switchboard SB-55( )/FTC, consisting of a nonmultiple, manually operated switchboard providing for two-way signalling and communication over all lines with the operator (see 8.1).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-S-571 — Solder; Soft (Tin, Tin-Lead, and Lead-Silver).

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing.)

LLL-F-291 — Fiberboard; Corrugated, Single Face (Flexible).

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corners.

PPP-B-636 — Boxes, Fiber.

PPP-T-60 — Tape, Pressure Sensitive Adhesive, for Packaging and Sealing.

MILITARY

MIL-P-75 — Packaging, Packing, and Container-Marking of Electron Tubes.

MIL-P-116 — Preservation, Methods of.

MIL-B-117 — Bags, Interior Packaging.

MIL-M-11473 — Magnetic Particle Inspection Soundness Requirements for Weldments.

FSC 5805

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MIL-S-11748 — Suppression Requirements for Electrical and Electronic Equipment (Except Internal Combustion Engine-Driven Equipment).

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-169 — Extreme-Temperature Cycle.

MIL-STD-170 — Moisture-Resistance Test Cycle for Ground Signal Equipment.

MIL-STD-252 — Wired Equipment, Classification of Visual and Mechanical Defects for.

DRAWINGS

SIGNAL CORPS

SC-DL-40463 — Telephone Switchboard SB-55( )/FTC Drawing and Data List.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N. W., Washington 6, D. C.)

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification

(Application for copies should be addressed to the Association of American Railroads, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Description. Telephone Switchboard SB-55( )/FTC is a nonmultiple, single position, common battery, manually operated switchboard. Operating on a 48-volt d.c. (direct current) supply it has a capacity for 100 fixed station lines and 2 trunk lines on a 3-cord circuit providing for two-way signalling and communication over all lines with the operator. The switchboard is floor mounted and is equipped with the following circuits:

Quantity	*Type of circuit*
100	Common battery manual line circuits.
2	Trunk circuits, dial and common battery manual.
3	Cord circuits.
1	Night alarm circuit.
2	Line-out alarm circuits.
1	Operator's circuit.
1	Alarm and ringing circuit.
1	Power-failure alarm circuit.
1	Heating circuit.

3.2 Construction. Telephone Switchboard SB-55( )/FTC shall be constructed in accordance with the drawings covered by Drawing and Data List SC-DL-40463.

3.3 Preproduction samples. The contractor shall furnish preproduction samples, for approval, as required by the invitation for bids and contract (see 6.2).

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3.4 **Parts, materials and processes: general.**

3.4.1 *Parts and materials.* Except as otherwise specified herein, parts and materials used in fabrication of the equipment shall be as specified in the drawings covered by Drawing and Data List SC-DL-40463.

3.4.2 *Processes.*

3.4.2.1 *Cleaning.* Metal parts after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled in the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.4.2.2 *Riveting.* After riveting, the joints shall be tight, the joined parts undamaged and the rivet heads properly seated and tight against their bearing surfaces.

3.4.2.3 *Soldering.*

3.4.2.3.1 *Solder.* Solder used for electrical connections shall be composition Sn60 conforming to QQ-S-571.

3.4.2.3.2 *Flux and cleaning agents.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux or soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.4.2.3.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The maximum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4.2.4 *Welding.* Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures.

3.4.2.4.1 *Cleaning.* Surfaces to be welded shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned of scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.4.2.4.2 *Process.* Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blister, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved. Inert-gas shielded arc welding shall be used, when practicable, for welding of aluminum or magnesium.

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3.4.2.4.3 *Nonvisual defects.* Welding, when subjected to the test of 4.10 shall evidence good fusion and shall be free of cracks, voids, and slag inclusions.

3.4.2.5 *Wiring and cabling.*

3.4.2.5.1 *Slack.* Sufficient slack shall be provided to prevent undue stress on cable forms, wires, and connections; to enable parts to be removed and replaced during servicing without disconnecting other parts; and to facilitate field repair of broken or cut wires.

3.4.2.5.2 *Protection.* Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes or slots in metal (partitions, shields, and similar items), they shall be protected by suitable grommets or bushings.

3.4.2.5.3 *Support.* Wires and cables shall be properly supported and secured, to prevent undue stress on conductors and terminals and undue change in position of the wires and cables after the equipment has been subjected to specified service conditions or has been serviced or repaired in a normal manner.

3.4.2.5.4 *Clearance.* Clearance between wires and cables, and parts, shall be sufficient to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards relays, or other parts, shall be at least ¼ inch unless use of specified parts makes such clearance impracticable.

3.4.2.5.5 *Connections.* Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Electrical connections shall not be made by clamping between a metallic and a nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of approved varnish.

3.4.2.5.6 *Terminals, fixed.* Fixed terminals (including binding posts) shall properly support the wires connected thereto and shall not turn or become loose when the equipment is subjected to specified service conditions.

3.5 **Electrical requirements.**

3.5.1 *Crosstalk.* All circuits shall be capable of meeting the test specified in 4.5.1.

3.5.2 *Insulation resistance.* The insulation resistance of all circuits shall be at least 100 megohms.

3.5.3 *Transmission loss.* Transmission loss between the switchboard terminals of any line or trunk and the operator's handset shall not exceed 2 db at a frequency of 1000 cycles per second and a power level of 10 dbm.

3.5.4 *Breakdown voltage.* Air gaps (in relays, switches, and similar items) and electrical insulation shall be capable of withstanding 500 volts without any visible deterioration or degradation in specified performance requirements. The test voltage shall be d.c. or peak a.c. of the same value and shall be applied for 10 seconds. If a.c., the waveform shall be sinusoidal and the frequency not over 100 cycles per second.

3.6 **Service conditions.** The equipment shall meet the following service conditions:

3.6.1 *Equipment nonoperating.* Except as stated in 3.6.1(b) the equipment shall comply with the operational requirements of 3.6.2 after subjection to any of the following non-operating conditions successively or in combination encountered during world-

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wide short-term storage and transit:

(a) *Temperature.* Exposure in the range of +160°F. to —80°F.; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours at any one time.

(b) *Relative humidity.* Relative humidity up to 97 percent, for an indefinite period of time; and exposure at 100 percent relative humidity, with condensation. for 4 hours. Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.6.2.

(c) *Orientation.* Indefinite storage in any orientation.

(d) *Bounce.* As specified in 3.7.

3.6.2 *Equipment operating.* After a warm-up period of 4 hours, the equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) *Temperature.* Ambient temperature in the range of +150°F. to —40°F. (The 150°F. temperature includes effect of sun-load.) Exposure at the high temperture extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours.

(b) *Relative humidity.* Relative humidity up to 97 percent, and exposure at 100 percent relative humidity, with condensation, for 4 hours.

(c) *Elevation.* Elevation up to 10,000 feet above sea level.

(d) *Orientation.* Any orientation up to 20° from normal operating orientation.

3.7 **Bounce.** The equipment shall meet full specification requirements after being subjected to the tests of 4.6.3.

3.8 **Preconditioning.** The equipment shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.9 **Radio interference supression.** The equipment shall conform to the requirements of Specification MIL-S-11748 and the interference limits set forth therein, with the antenna of the interference measuring equipment located 2 feet from the equipment.

3.10 Interchangeability. Corresponding units and replaceable assemblies, subassemblies, and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other equipment with which the items are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular subassembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.11 Serial numbers. Telephone Switchboard SB-55( )/FTC shall be serial numbered (see 6.2).

3.12 **Workmanship.** The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.4.2.1 *Cleaning.*

3.4.2.2 *Riveting.*

3.4.2.3 *Soldering.*

3.4.2.4 *Welding.*

3.4.2.5 *Wiring and cabling.*

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4. QUALITY ASSURANCE PROVISIONS

4.1 **Classification of inspection.** Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery).

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified in 4.3 and 4.12.)

(1) Procurement inspection of items before preparation for delivery (see 4.3).

(2) Procurement inspection of preparation for delivery (see 4.12 and section 5).

4.2 **Preproduction inspection.** This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, and the inspection items listed in table II. Statistical sampling shall not be a part of preproduction inspection. Other nondestructive inspection may be performed on preproduction sample to determine compliance with specified requirements.

TABLE I. *Preproduction inspection*

Inspection	Requirement paragraph	Inspection paragraph
Temperature	3.6	4.6.2
Moisture resistance	3.6	4.6.1
Bounce	3.7	4.6.3
Operational	...	4.7
Radio interference	3.9	4.8
Interchangeability	3.10	4.9
Welding	3.4.2.4	4.10
Workmanship	3.12	4.11

4.3 **Procurement inspection of items before preparation for delivery.** The minimum inspection performed by the contractor shall be as specified by (a) through (c) below to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government — at its discretion — will verify, by product inspection the contractor's compliance with specified requirements. Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of the group A inspection specified in 4.3.1.

(b) When MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Each unit which will be subjected to the group A inspection shall be preconditioned after final assembly (see 3.8).

4.3.1 *Group A inspection.* This inspection (including sampling) shall conform to table II and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational test shall be last.

4.3.2 *Disposition of nonconforming product.* When defective sample units or failed inspection lots are submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

4.4 **Bounce preconditioning.** The equipment shall be placed in its normal operating position on the table of the package tester as made by L. A. B. Corporation, Summit, N. J., or equal. The package tester, shafts in phase, shall have a speed such that it is

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TABLE II. *Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Electrical	3.5	4.5	40 dphu	
Crosstalk	3.5.1	4.5.1	for	
Insulation resistance	3.5.2	4.5.2	the	
Transmission loss	3.5.3	4.5.3	group	
Breakdown voltage	3.5.4	4.5.4	...	
Operational	...	4.7	1.0 %	
Visual and mechanical	3.12	4.11	40 dphu	15 dphu

just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute. After bounce preconditioning the equipment shall not be repaired, alined, cleaned or otherwise changed prior to subjection to acceptance inspection.

4.5 Electrical tests.

4.5.1 *Crosstalk.* Each trunk, line, and cord circuit on the switchboard shall be tested for crosstalk coupling with every other circuit, as follows:

(a) Connect 600-ohm loads across the two circuits being tested.

(b) Apply 2.0 volts r.m.s. at 1000 cycles per second across one of the two circuits.

(c) The voltage at any point across the other circuit shall not exceed one millivolt.

4.5.2 *Insulation resistance.* The insulation resistance of each circuit shall be measured with a potential of not less than 100 volts d.c. to determine compliance with 3.5.2.

4.5.3 *Transmission loss.* The switchboard shall be tested for compliance with 3.5.3. The transmission loss shall be tested on each trunk circuit and on ten of the 100-line circuits, selected at random.

4.5.4 *Breakdown voltage.* The equipment shall be tested to determine compliance with 3.5.4.

4.6 Service condition tests.

4.6.1 *Moisture resistance test for equipment.*

4.6.1.1 *Test conditions.*

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.6.1.2 *Test method.*

(a) Dry at 130° ±5°F. for 24 hours.

(b) Condition at 77° ±5°F. and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.6.1.3 and re-adjust or re-aline as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements

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specified in 4.6.1.3 shall be made during the periods shown on the standard.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F. and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specification requirements for those measurements specified in 4.6.1.3.

4.6.1.3 *Performance.* Insulation resistance during and after the test, measured at a potential of 500 volts, shall be at least 50 megohms between any two circuits or between any circuit and the frame. The equipment shall be capable of passing the operational test of 4.7 after completion of the test.

4.6.1.4 *Failure.* If any equipment fails to meet the performance specified in 4.6.1.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.6.1.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.6.1.3, it does not pass the test.

4.6.2 *Temperature.* The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. The tests of 4.5 and 4.7 shall be performed at step 3 and at step 9.

4.6.3 *Bounce test.* The equipment shall be placed on the table of the package tester as made by the L. A. B. Corporation, Summit, N. J., or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 r.p.m. ±1.0 percent, for a total of 3 hours. During each ½ hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.7 **Operational test.**

(a) Connect the equipment to a line.

(b) Check ringing, transmission, and supervision of each cord.

(c) The S (supervisory) lamp should flash when the switchboard is operated and released.

(d) The LO (line out) lamp should flash when the line is opened and closed.

(e) Check operation of the night alarm, LO alarm, ringing signal lamp, power failure alarm, hand generator, and generator switch.

(f) Call in from the telephone connected to the line being tested, through the test equipment, thus lighting the line camp.

(g) Operate the hookswitch and check to see that the line lamp goes on and off.

(h) Answer to call, check quality of transmission, hang up, then ring on the line.

(i) See that the annunciator lamp lights.

(j) Operate the annunciator release key, extinguishing the annunciator lamp.

(k) Flash the LO lamp.

(l) Check the covers of the A and LO relays to make certain they do not touch the contacts of the relays.

(m) Check fuse posts for crosses with each other and grounds.

(n) Check continuity from ground to

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relay mounting strips, switch frame, and jack frame.

(o) With trunk and cord switches normal, check to make certain that the P terminal of the 72-EA induction coils is not grounded.

4.8 **Radio interference suppression test.** The equipment shall be tested for compliance with 3.9.

4.9 **Inspection for interchangeability.** Replaceable parts, subassemblies and components shall be inspected to determine compliance with the physical interchangeability requirements of 3.10. Replaceable parts, subassemblies and components which do not meet the requirements of 3.10 shall be considered a major defect.

4.10 **Welding.** Welding shall be inspected by the magnetic particle method specified in Specification MIL-M-11473, and by the ultrasonic method using a reflectoscope as made by Sperry Products, Incorporated, Danbury, Conn., or equal. Both tests shall be performed on each sample unit.

4.11 **Visual and mechanical inspection.** The equipment shall be examined for the defects listed in Standard MIL-STD-252 and to determine compliance with the requirements of 3.12.

4.12 **Procurement inspection of preparation for delivery.** The minimum inspection performed by the contractor shall be as specified in 4.12.1 and 4.12.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government — at its discretion — will verify, by product inspection, the contractor's compliance with specified requirements (see 6.11). Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.12.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall conform to tables III and IV.

4.12.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed in table III and table IV and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be 4 percent and the AQL for minor defects shall be 10 percent. Classification of defects shall be as shown in tables III and IV.

4.12.3 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.12.4 *Procedure in case of failure.* If an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and the Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.12.5 *Disposition of nonconforming product.* Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of

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rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

TABLE III. *Classification of defects for preservation, packaging, and marking*

Major defects

1. Use of improper or defective material
2. Quantity in unit package not as specified.
3. Incorrect packaging method applied.
4. Cushioning or padding omitted.
5. Cushioning inadequate for the physical and mechanical protection of the item.
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.
7. Stock number omitted, incorrect, or illegible.
8. Nomenclature omitted, incorrect, or illegible.
9. Marking of quantity of items in package omitted, incorrect, or illegible.
10. Different stock numbered items in the same unit package.

Minor defects

1. Item not properly blocked or braced within the unit package to prevent movement.
2. Packaging material damaged.
3. Air not expelled from barrier.
4. Conforming wraps are not snug-fitting and contain voids.
5. Any item of marking information other than 7 through 9 omitted, incorrect, or illegible.

TABLE IV. *Classification of defects for packing and marking*

Major defects

1. Use of improper or defective material.
2. Quantity in pack not as specified.
3. Gross weight in excess of specified amount.
4. Box closure not as specified.
5. Type, grade, class, and style of the shipping container not as specified.
6. Strapping omitted.
7. Strapping inadequate or incorrectly applied.
8. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
9. Shipping documents or packing list omitted.
10. Stock number omitted, incorrect, or illegible.
11. Nomenclature omitted, incorrect, or illegible.
12. Marking of quantity of items in pack omitted, incorrect, or illegible.
13. Destination marking omitted, incorrect, or illegible.
14. Service designation (color marking) omitted.
15. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.
16. Overseas code marking omitted, incorrect, or illegible.

Minor defects

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Any item of required marking information other than 10 through 16 listed under major defects omitted, incorrect, or illegible.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Telephone Switchboard SB-55( )/FTC shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Military Specification MIL-P-116 and described herein.

5.1.1.1 *Technical manual.* Each technical manual shall be packaged method IC-3 as follows: Inclose each technical manual with-

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in a close-fitting, type II, class b bag, conforming to Military Specification MIL-B-117. Heat seal to effect closure.

5.1.1.2 *Tubes (spare).* Spare tubes shall be packaged individually in accordance with the applicable provisions of Military Specification MIL-P-75.

5.1.1.3 *Lamps (spare).* Spare lamps shall be packaged method III, as follows: Place spare lamps within a folding, paperboard carton, conforming to Federal Specification PPP-B-566. Separate each lamp by fitting the carton with partitions. Place each lamp within the partitions. Properly seal closure.

5.1.1.4 *Fuses (spare).* Spare fuses shall be packaged method III, as follows: Fuses bearing the same stock number shall be placed together within a wrap of flexible, single-face, corrugated fiberboard, type I, conforming to Federal Specification LLL-F-291. Secure the wrap with gummed paper tape, class 1 conforming to Federal Specification UU-T-111.

5.1.1.5 *Consolidated package.* Consolidate the items, packaged as specified in 5.1.1.2 through 5.1.1.4, together within a close-fitting, type I, class 1, RSC box, conforming to table I of Federal Specification PPP-B-636. Closure shall be made in accordance with the appendix of the box specification.

5.1.1.6 *Switchboard SB-55( )/FTC.* Each switchboard shall be packaged method IC-5, as follows: Secure the consolidated package within the switchboard frame. Block and brace all loose or movable equipment within the cabinet with cells or pads or both, fabricated of double-faced, corrugated fiberboard, having a minimum dry bursting strength of 175 pound, designed to absorb the shock if impact encountered in handling and transit. Close cabinet and secure all fastenings. Fabricate a corrugated fiberboard, 275-pound test blocking assembly, to bridge the unit and square same up for proper packaging; material shall conform to table II of Specification PPP-B-636. Cushion each switchboard on the four corners having the longest dimensions with double-fold pads, fabricated of "A" or "C" flute corrugated fiberboard, having a minimum dry bursting strength of 200 pounds, conforming to table II of Federal Specification PPP-B-636. Fabricate a corrugated fiberboard sleeve, extending the full length of the cabinet, with two end caps (one for each end), of regular half-slotted containers, to completely inclose the cabinet (see fig. 1), fabricated of corrugated fiberboard, 400-pound test, V3c, conforming to table II of Federal Specification PPP-B-636. Closure shall be in accordance with the applicable requirements of the Appendix of the box specification. In addition, seal all seams and joints with 3-inch-wide waterproof, pressure-sensitive tape, type III, class 3, conforming to Federal Specification PPP-T-60.

5.1.2 *Level C.* Switchboard SB-55( )/FTC shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

5.2 Packing.

5.2.1 *Level A.* Each switchboard SB-55 ( )/FTC shall be packed in a Class 2, Style 2, nailed, wood box, conforming to Federal Specification PPP-B-621, modified to include a skid-type base and interior diagonal bracing (see fig. 2). The size of the cleats and interior bracing members shall be determined by limitations of weight and volume. The sheathing parallel to the skids shall run vertically to permit nailing into the skids. Fabricate the box to fit the contents snugly. Place the technical manual packaged as specified in 5.1.1.1, between the packaged item and the lid of the shipping

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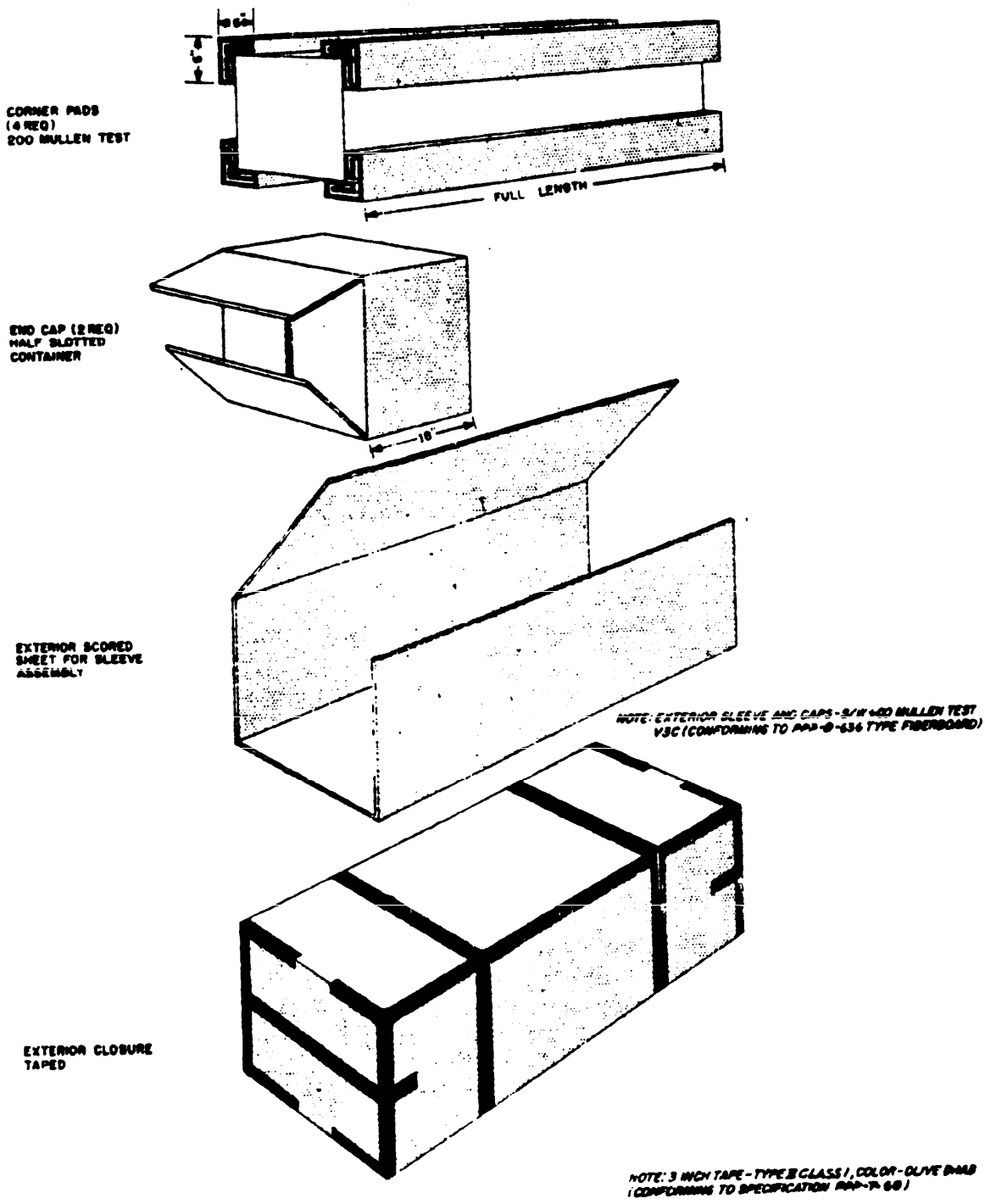
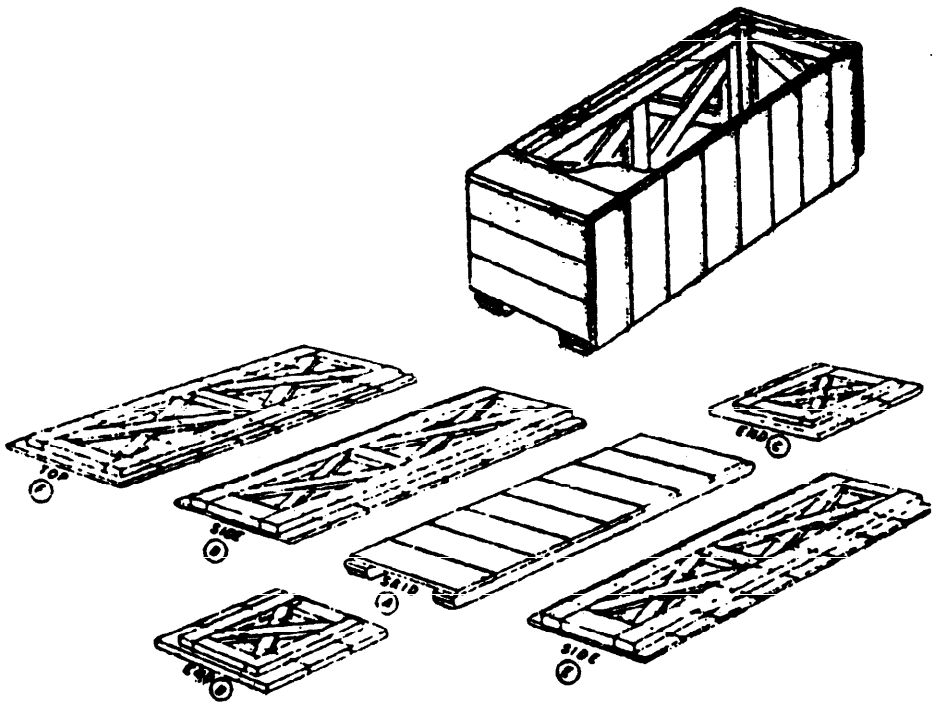


FIGURE 1.

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MODIFIED STYLE 2

"XX" PANEL CONSTRUCTION



1. The skid Ⓐ sides Ⓑ Ⓔ ends Ⓕ Ⓒ and top Ⓓ shall be prefabricated as independent units as illustrated.

2. Assemble prefabricated units as follows.

NAIL Ⓓ & Ⓔ TO Ⓐ
NAIL Ⓕ & Ⓒ TO ⒶⒹ & Ⓔ
NAIL Ⓔ TO ⒷⒸⒹ Ⓔ

FIGURE 2.

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container. Closure shall be in accordance with the box specification.

5.2.1.1 *Metal strapping.* Shipping containers shall be strapped with flat steel strapping, in accordance with the requirements of the appendix of the container specification, only for direct shipment to ports.

5.2.2 *Level B.* Each Switchboard SB-55 ( )/FTC shall be packed as specified in 5.2.1, except that the modified Style 2, nailed, wood box shall be Class 1, conforming to Federal Specification PPP-B-621.

5.2.3 *Level C.* Switchboard SB-55( )/FTC shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

5.3 **Marking.** Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 **Intended use.** Switchboard, Telephone, Manual (Telephone Switchboard SB-55( )/FTC) is intended for use in fire-reporting telephone systems of permanent installation in the base station of a communication zone.

6.2 **Ordering data.** Procurement documents should specify the following:

6.2.1 *General.*

(a) Title, number, and date of this specification.

(b) Whether domestic or overseas shipment is required.

(c) Quantity of tools and running spare parts required.

(d) Technical literature required.

(e) Facsimiles required.

(f) Serial numbers.

(g) Preproduction samples required.

6.2.2 *Preparation for delivery.*

(a) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(b) Preproduction pack(s) as follows:

(1) Makeup of pack(s).

(2) Number of each kind of pack to be submitted.

(3) Inspection that will be performed thereon—including rough handling test, which will not be performed as acceptance inspection.

(4) Five copies of process sheet, using format shown on figure 3, and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

(5) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

6.3 **Nomenclature.** The parentheses in the nomenclature will be deleted and replaced by a letter identifying the particular design, for example SB-55W/FTC. As one as pos-

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sible after award of the contract, the contractor should apply to the contracting officer for such information.

6.4 Definitions.

6.4.1 *Inspection.* Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.2 *Examination.* Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.4.3 *Testing.* Testing consists of determinations of compliance, using technical means.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Air Force
Navy—Bureau of Yards and Docks

Other interest:
Army—O

Preparing activity:
Army—Signal Corps

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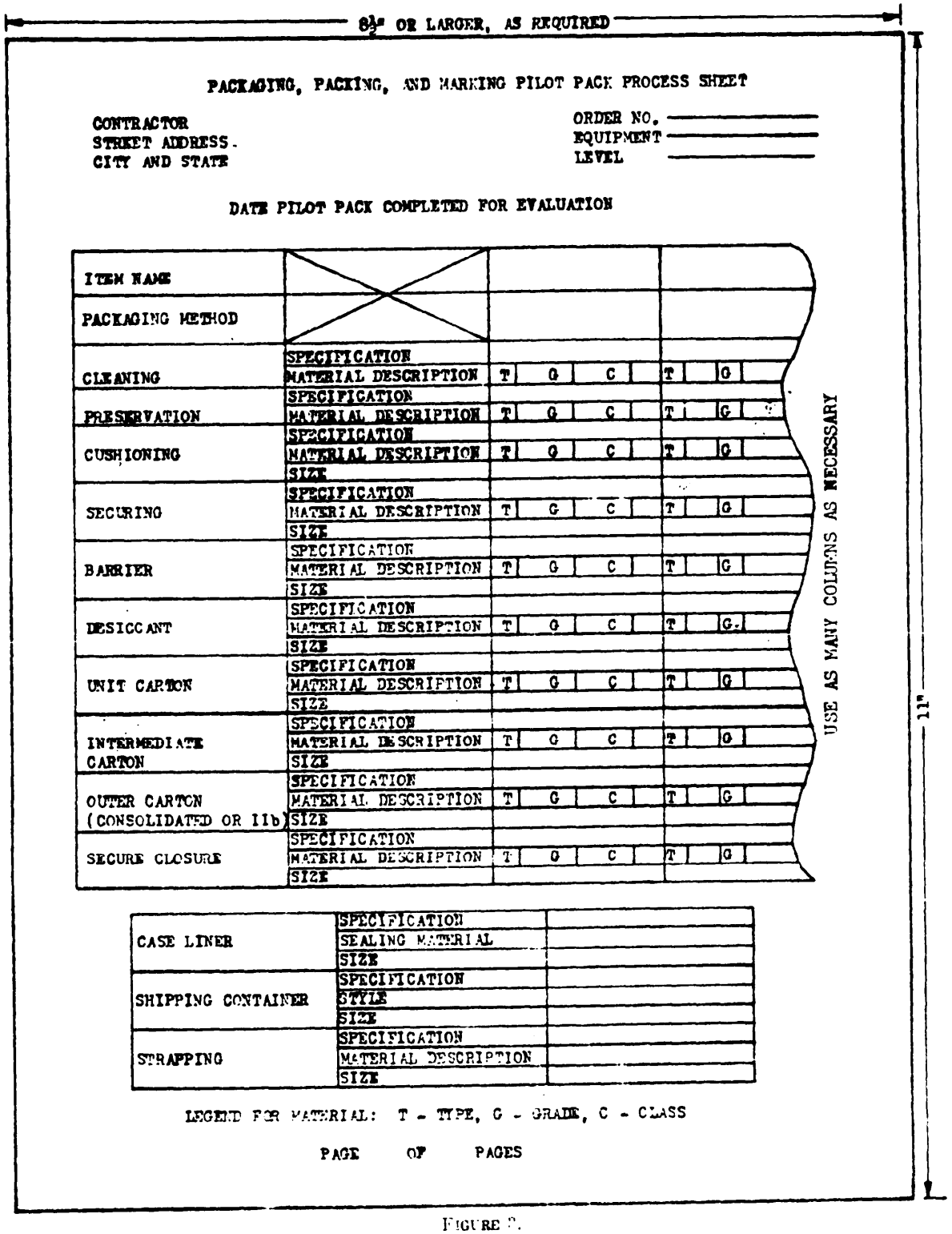


FIGURE 2.

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